Curating Multi-Mode CoT for Efficient Math Reasoning with GPT-OSS

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Abstract. Knowledge distillation transfers reasoning capabilities from large teacher models to smaller student models through various approaches, including mimicking intermediate representations, probability matching, or curating teacher-generated reasoning traces datasets. In a mathematical context, dataset-based distillation is often limited by noisy verbose supervision. GPT-OSS, a series of OpenAI state-of-the-art open-weight language models, exposes explicit low/medium/high inference modes, offering a controllable source of reasoning at different "effort" levels. We present a straightforward pipeline to generate multi-mode CoT from GPT-OSS and curate it into mode-specific supervision for math question answering. Our curation applies two steps, including final-answer verification to remove incorrect samples, followed by length distribution-based filtering and median-length correct trace selection to mitigate excessive verbosity. Experiments on GSM8K and MATH500 with Llama 3.2 3B show that curated mode-specific SFT outperforms training on raw teacher outputs and provides strong initialization for GRPO, reaching 80.06% GSM8K accuracy (0-shot) when initialized from curated low-mode SFT. Overall, quality- and length-controlled reasoning supervision is more effective than unfiltered CoT quantity for small-model math QA. Our code and datasets are publicly available at https://github.com/Koii2k3/LLaMA-OSS.

Keywords: Chain-of-thought · Knowledge distillation · Data curation.

1 Introduction and Related Works

Question answering (QA) is a long-standing core task in natural language processing, where a system must map a natural language question to a correct answer. For tasks that demand reasoning capabilities, especially math, coding,

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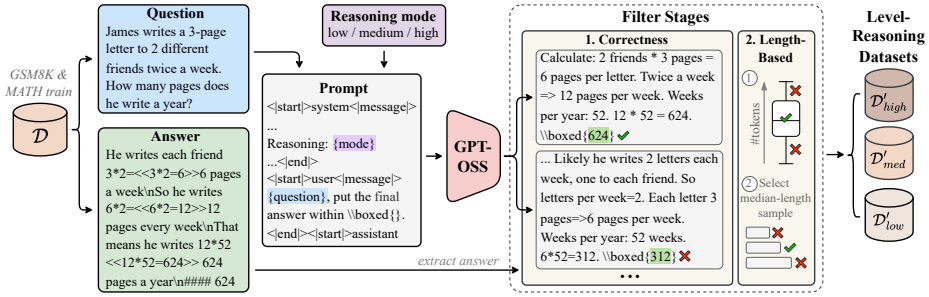

Fig. 1: Overall pipeline of the multi-mode CoT generation and curation pipeline.

and scientific QA, these tasks become more challenging and require multi-step reasoning to reach an exact final answer [3]. Although large language models (LLMs) achieve strong results on math benchmarks, their cost motivates transferring reasoning ability into smaller, more efficient models.

A common approach is to distill the chain-of-thought (CoT) and predictions generated by a teacher model as new training data for a smaller model via sampling selection [8], followed by supervised fine-tuning (SFT) and, in certain settings, reinforcement learning, for example, group relative policy optimization (GRPO), with task rewards [2]. However, raw CoT generated from high-capacity teacher model is often noisy, inconsistent between samples, and highly variable in length, which can degrade small-model training [9]. GPT-OSS [1], which is a family of open-weight, text-only reasoning models that supports low/medium/high "reasoning effort" modes, enables controllable "effort-level" reasoning generation. We leverage this feature and propose a pipeline to curate multi-mode GPT-OSS output into clean supervision datasets for math QA.

In this paper, our contributions can be summarized as follows:

1. We propose a reproducible pipeline to construct reasoning-level QA datasets $\mathcal{D}_{\text{low}}, \mathcal{D}_{\text{med}}, \mathcal{D}_{\text{high}}$ from the original datasets utilizing GPT-OSS.
2. We conduct an empirical study showing that fine-tuning with the filtered reasoning-level datasets enables a 3B Llama 3.2 model to achieve better performance on math benchmarks.

The remainder of this paper is organized as follows. Section 2 details our data curation methodology, Section 3 presents the experimental results, and Section 4 concludes the paper.

2 Methodology

2.1 Problem Formulation

Let $\mathcal{D} = \{(q, a)\}$ denote the union of the GSM8K [3] and MATH [6] training splits, where q is a question, and a is the ground-truth final answer. As illustrated in Fig. 1, reasoning-level datasets are constructed from $\mathcal{D}$ using GPT-OSS and a two-stage filtering procedure.

2.2 Reasoning-level Dataset Generation

GPT-OSS is used to generate responses for each problem in three reasoning modes $m \in \{\text{low}, \text{med}, \text{high}\}$. For each mode, we generate $K = 5$ candidate responses with mode-specific system prompts, yielding 15 total responses per problem. Each response r follows the format $r =$ `<think>` π `</think> \boxed{`$\hat{a}$`}`, where π denotes the reasoning trace and $\hat{a}$ is the predicted final answer, while `\boxed{}` provides a unified delimiter for programmatic extraction across both datasets. This produces the raw dataset $\mathcal{D}^* = \{(q, a, r_{\text{low}}, r_{\text{med}}, r_{\text{high}})\}$.

The original reasoning dataset is denoted by $\mathcal{D}_{\text{orig}} = \{(q, a, r_{\text{orig}})\}$, where r_{orig} is constructed from the dataset-provided reasoning trace π_{orig} and the ground-truth final answer a in the same response format.

2.3 Two-Stage Filtering

The raw generations in $\mathcal{D}^*$ are filtered to obtain mode-specific datasets $\mathcal{D}'_{\text{low}}$, $\mathcal{D}'_{\text{med}}$, $\mathcal{D}'_{\text{high}}$ to ensure that the SFT process imitates only accurate reasoning logic while strictly adhering to the desired reasoning density:

- **Correctness Filtering.** We retain only responses whose extracted answer matches the ground truth to prevent the distillation of incorrect reasoning patterns or logical inconsistencies into the student model:

$$\mathcal{R}'_m = \{(q, a, r) \mid \text{extract}(r) = a\}, \tag{1}$$

  where extract$(\cdot)$ parses the predicted answer $\hat{a}$ from the `\boxed{}` format.
- **Length-Based Filtering.** To ensure that the depth of reasoning is aligned with the selected mode while excluding overly terse or verbose responses, we propose length distribution-based filtering of generated traces. Let $\ell(r)$ be the token length of reasoning r, computed via the GPT-OSS tokenizer. For each dataset $d \in \{\text{GSM8K}, \text{MATH}\}$ and mode m, we compute quartiles $Q_{25}^{d,m}$ and $Q_{75}^{d,m}$ from $\{\ell(r) \mid r \in \mathcal{R}'_m\}$, retaining responses satisfying $Q_{25}^{d,m} \leq \ell(r) \leq Q_{75}^{d,m}$. When multiple valid responses exist for the same problem and mode, we retain the response closest to the median reasoning length as a stabilizer:

$$r^* = \arg\min_{r \in \mathcal{R}_q} \left| \ell(r) - \text{median}_{r' \in \mathcal{R}_q} \ell(r') \right|, \tag{2}$$

  where $\mathcal{R}_q$ denotes the retained responses for problem q.

2.4 Reasoning-level Dataset and Training

After filtering, three datasets are obtained, which are later used for SFT. $\mathcal{D}'_{\text{low}}$ contains 8,013 samples, $\mathcal{D}'_{\text{med}}$ contains 7,880 samples, and $\mathcal{D}'_{\text{high}}$ contains 8,174 samples. Each dataset comprises a balanced mixture of GSM8K and MATH problems with mode-specific reasoning depths to instill distinct behavioral identities across reasoning levels.

3 Experiments and Results

3.1 Experimental Setup

Models and Datasets. The 20B open-weights GPT-OSS model is employed as the teacher generator to produce reasoning trajectories, with its behavior modulated via system-level prompts. The student model is Llama 3.2 3B Instruct [4], serving as the target model. Our training corpus comprises the training splits of GSM8K [3] and MATH [6], consisting of 7,473 grade-school and 7,500 competition-level problems, respectively.

Training and Evaluation. SFT uses question-response pairs from various datasets, such as (q, r^*) from each curated $\mathcal{D}'_m$, (q, r_{orig}) from $\mathcal{D}_{orig}$, and (q, r_m) from all of $\mathcal{D}^*$. For GRPO, policy optimization is applied to (q, a) pairs from $\mathcal{D}$ to generate new reasoning trajectories, which are refined with reward-based gradients. Evaluation reports accuracy in the 0-shot setting, where the model solves questions without examples, and the few-shot setting, where the model is given several solved examples before each test question. We use 5-shot prompting for GSM8K and 4-shot for MATH500 [7], following standard practice.

Implementation. Experiments are conducted on a single NVIDIA RTX A5000 GPU, and the maximum generation length is set to 2,048 tokens. SFT runs for one epoch using LoRA with rank 16 and alpha 32, a global batch size of 32, and a learning rate of 2×10^{-5}. GRPO generates 8 responses per prompt with temperature $T = 1.0$ and top-$p = 0.95$, and training runs for 300 steps with a learning rate of 1×10^{-5}. LoRA is applied with rank 8 and alpha 32. The reward uses binary correctness $r_{\text{acc}} \in \{0, 1\}$ and a 5-gram repetition penalty whose minimum value is -0.5.

3.2 Results

Table 1 compares SFT and a two-stage setting where SFT is followed by GRPO. Two consistent patterns are observed. First, curation quality determines whether SFT improves or degrades performance. Second, GRPO yields the largest gains when initialized from a strong and stable SFT checkpoint.

Impact of Data Curation. SFT on curated reasoning-level datasets $\mathcal{D}'_{\text{low}}$, $\mathcal{D}'_{\text{med}}$, and $\mathcal{D}'_{\text{high}}$ improves performance over SFT on the original dataset, the raw dataset $\mathcal{D}^*$ and the baseline model as well. Specifically, SFT on the raw-generated dataset $\mathcal{D}^*$ reduces accuracy across all evaluation settings, indicating that unfiltered teacher traces provide detrimental supervision. Among curated variants, $\mathcal{D}'_{\text{low}}$ yields the most stable improvements, reaching **0.7111** on GSM8K in the 0-shot setting and **0.7142** in the 5-shot setting. The model trained on $\mathcal{D}'_{\text{high}}$ achieves the best SFT result on MATH500 0-shot at **0.4140**, yet its few-shot performance is less consistent. Overall, the results suggest that concise and

Table 1: Performance of Llama 3.2 3B across SFT and GRPO stages. Model suffixes denote the SFT checkpoint used for GRPO initialization. Bold indicates best results per section.

Model	Training	Dataset	GSM8K		MATH500	
			0-shot	5-shot	0-shot	4-shot
Llama3.2	-	-	0.7043	0.7043	0.3960	0.4340
	SFT	$\mathcal{D}_{orig}$	0.7043	0.7104	0.4060	0.4240
		$\mathcal{D}^*$	0.6876	0.5762	0.3400	0.2420
		$\mathcal{D}'_{\text{low}}$	**0.7111**	**0.7142**	0.4100	**0.4400**
		$\mathcal{D}'_{\text{med}}$	0.7051	0.7074	0.4000	0.4160
		$\mathcal{D}'_{\text{high}}$	0.7013	0.7051	**0.4140**	0.3920
Llama3.2	GRPO	$\mathcal{D}$	0.7771	0.6603	0.4540	0.4380
Llama3.2-$\mathcal{D}_{orig}$			0.7847	0.6156	0.4520	0.4560
Llama3.2-$\mathcal{D}^*$			0.6308	0.5861	0.4400	0.4220
Llama3.2-$\mathcal{D}'_{\text{low}}$			**0.8006**	0.7195	**0.4760**	0.4520
Llama3.2-$\mathcal{D}'_{\text{med}}$			0.7771	0.6323	0.4480	**0.4600**
Llama3.2-$\mathcal{D}'_{\text{high}}$			0.7559	**0.7225**	0.4740	**0.4600**

verified traces are distilled more effectively into a 3B model than longer and more complex traces.

Effectiveness of GRPO. We observe that the GRPO gain depends on the SFT starting point. The best 0-shot results occur when GRPO is initialized from the SFT checkpoint trained on $\mathcal{D}'_{\text{low}}$, reaching **0.8006** on GSM8K and **0.4760** on MATH500, indicating that SFT on curated traces provides reliable priors for a more effective policy optimization on $\mathcal{D}$. In few-shot settings, the best GSM8K 5-shot score is achieved when GRPO is started from $\mathcal{D}'_{\text{high}}$ at **0.7225**, while the best MATH500 4-shot score is shared by initializations of $\mathcal{D}'_{\text{med}}$ and $\mathcal{D}'_{\text{high}}$ at **0.4600**. In general, SFT controls reasoning style and stability, whereas GRPO primarily refines answer correctness through reward-driven updates, consistent with prior findings on reinforcement learning for reasoning, such as ThinkDial [5].

3.3 Ablation Study

We investigate whether mixing reasoning-level datasets improves generalization in Table 2. It should be noted that fine-tuning with merged datasets such as $\mathcal{D}'_{\text{low}} \cup \mathcal{D}'_{\text{med}}$ reaches **0.7066** and outperforms that with $\mathcal{D}'_{\text{high}}$ at **0.7013**, yet still does not exceed the best single-mode setting on GSM8K at **0.7111** on $\mathcal{D}'_{\text{low}}$. We hypothesize that directly mixing reasoning styles with mismatched token length distributions and internal logic creates interference, preventing the small model from internalizing a coherent policy. This suggests that effective multi-mode distillation requires a structured curriculum, such as an easy-to-hard approach, rather than random mixing, which is also consistent with [10].

Table 2: SFT performance comparison of multi-mode training combinations.

Model	Dataset	GSM8K 0-shot	MATH500 0-shot
Llama3.2	$\mathcal{D}'_{\text{low}} \cup \mathcal{D}'_{\text{med}}$	0.7066	<u>0.3980</u>
	$\mathcal{D}'_{\text{med}} \cup \mathcal{D}'_{\text{high}}$	0.6975	**0.4080**
	$\mathcal{D}'_{\text{low}} \cup \mathcal{D}'_{\text{high}}$	<u>0.7081</u>	<u>0.3980</u>
	$\mathcal{D}'_{\text{low}} \cup \mathcal{D}'_{\text{med}} \cup \mathcal{D}'_{\text{high}}$	**0.7096**	0.3920

4 Conclusion and Future Work

In this work, we proposed a straightforward yet effective data curation pipeline that generates diverse trajectories under three reasoning modes (low, medium, high) and filters them into clean reasoning-level datasets with benchmarking on math QA datasets. Our method and experiments highlighted that, to improve a model's reasoning ability, the quality of reasoning traces is often more important than quantity during fine-tuning, especially for small model distillation. Based on our GRPO experiments and ablation results on mode combinations, one limitation and two future directions are identified. First, the proposed curation relies on reasoning length without explicitly evaluating the logical quality of generated traces. Future work can incorporate quality-aware signals under adaptive reward functions within GRPO. Second, a structured, easy-to-hard curriculum could be explored to reduce interference between reasoning styles.

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